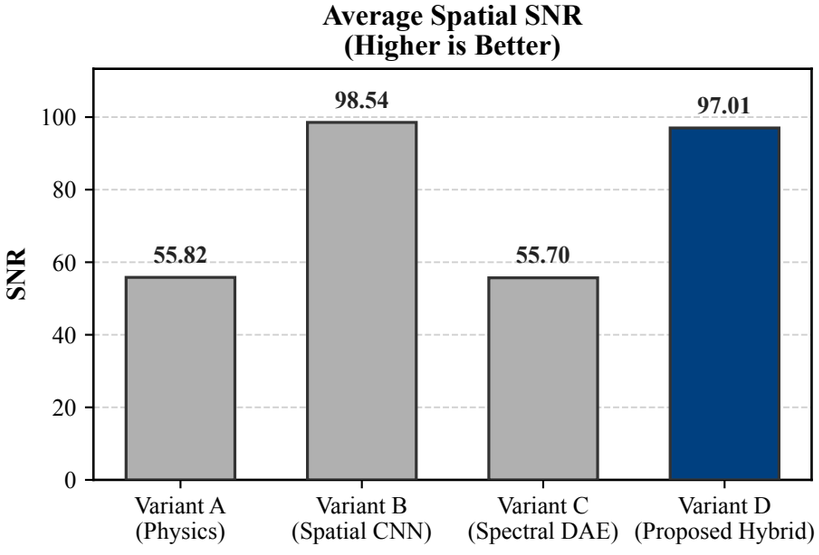
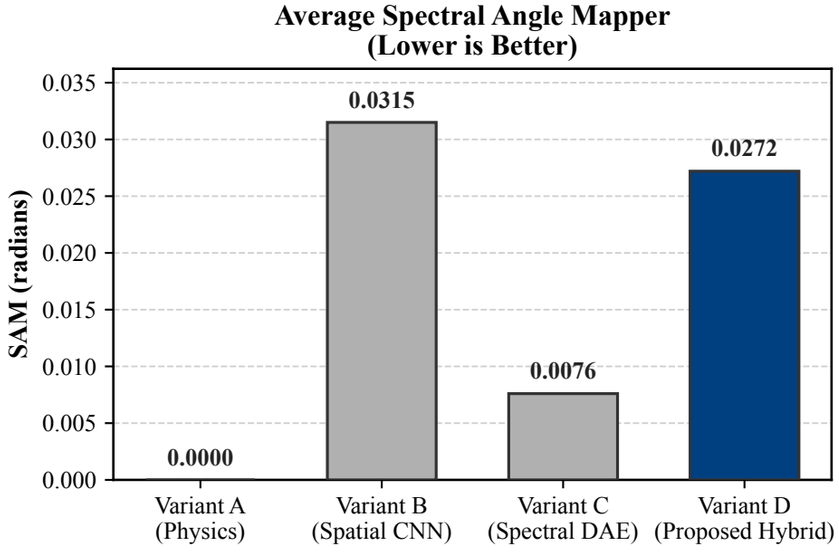
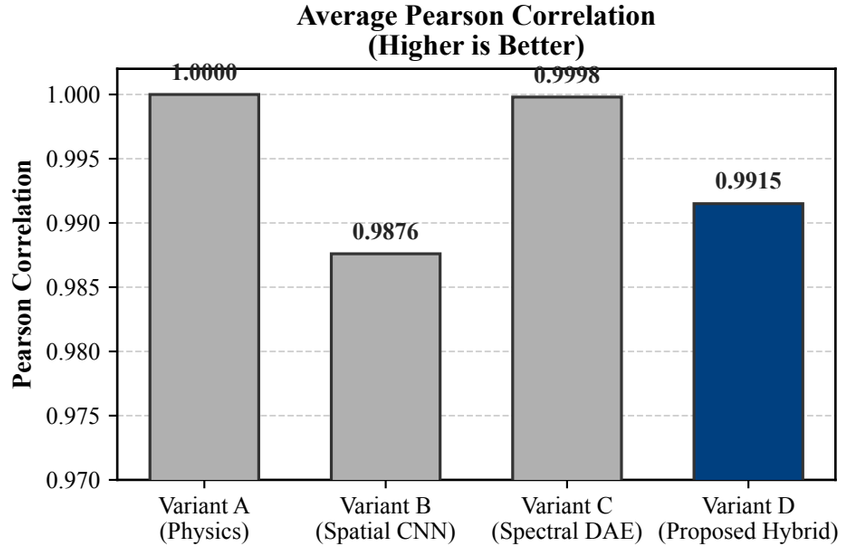
Ablation Analysis of the Proposed Spatial–Spectral Hybrid Denoising Pipeline







Key Findings

- Variant A represents the physics-based correction stage and serves as the reference.
- Variant B demonstrates that the Spatial CNN is primarily responsible for spatial stripe suppression and SNR improvement.
- Variant C shows that the Spectral Noise2Void autoencoder primarily preserves spectral fidelity with minimal spatial enhancement.
- Variant D combines both modules, maintaining nearly all spatial enhancement while improving spectral fidelity through reduced SAM and higher Pearson correlation.

** Variant A is the reference stage; therefore, SAM = 0 and Pearson correlation = 1 by definition because the physics-corrected image is compared with itself.*